

# context\_compactor

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## context\_compactor.sh – Companion Guide

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### Overview

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§11.4.141 M7 implementation. Prunes stale tool-results and old thinking from long sessions. NEVER touches the governance prefix.

### Functions

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| Function                               | Purpose                            | Example                                |
|----------------------------------------|------------------------------------|----------------------------------------|
| <code>cc_should_compact</code>         | Check if context exceeds threshold | <code>cc_should_compact 50</code>      |
| <code>cc_prune_stale_results</code>    | Mark old results as prunable       | <code>cc_prune_stale_results 20</code> |
| <code>cc_summarize_old_thinking</code> | Compress old thinking blocks       | <code>cc_summarize_old_thinking</code> |
| <code>cc_preserve_governance</code>    | Documented invariant (no-op)       | <code>cc_preserve_governance</code>    |
| <code>cc_get_context_stats</code>      | Return context statistics          | <code>cc_get_context_stats</code>      |

## Usage

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```
source constitution/scripts/token_efficiency/context_compactor.sh
```

```
# Check if compaction needed (after 50 turns)
```

```
if cc_should_compact 50; then
```

```
    cc_prune_stale_results 20
```

```
    cc_summarize_old_thinking
```

```
fi
```

```
# Governance preservation is automatic (invariant)
```

## Invariant

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The governance prefix (CLAUDE.md, Constitution.md) is NEVER touched by compaction. It is the cache breakpoint and must remain byte-stable for prompt caching.

## Cross-references

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- Constitution §11.4.141 M7
- `token_efficiency_lib.sh` — for status and caching
- `scoped_read.sh` — for targeted reads